

SMART LIGHTING SYSTEM USING LDR, LCD & ARDUINO UNO

TASK-2 REPORT

Submitted By: Enagandla Thrisha

BVRIT - EEE Department

INTRODUCTION

Smart Lighting System

A Smart Lighting System is an automated lighting solution that controls lights based on environmental conditions without requiring manual intervention. It uses sensors, microcontrollers, and control algorithms to improve energy efficiency and user convenience.

In this project, a Light Dependent Resistor (LDR) is used to detect the surrounding light intensity. The Arduino Uno continuously monitors the light level and automatically controls an LED. When the environment becomes dark, the LED turns ON, and when sufficient light is available, the LED turns OFF. The system status and light intensity are displayed on an LCD screen.

Smart lighting systems are widely used in street lighting, smart homes, industrial facilities, parking areas, and security systems to reduce power consumption and improve automation. By automatically adjusting lighting based on ambient conditions, these systems contribute to energy conservation and efficient resource utilization.

Objective

To design and simulate a Smart Lighting System using Arduino Uno, LDR sensor and LCD display that automatically controls lighting based on ambient light conditions.

Components Required

- Arduino Uno
- LDR Sensor Module
- LED
- 220 Ω Resistor
- 16x2 I2C LCD
- Jumper Wires
- Wokwi Simulator

CIRCUIT DIAGRAM

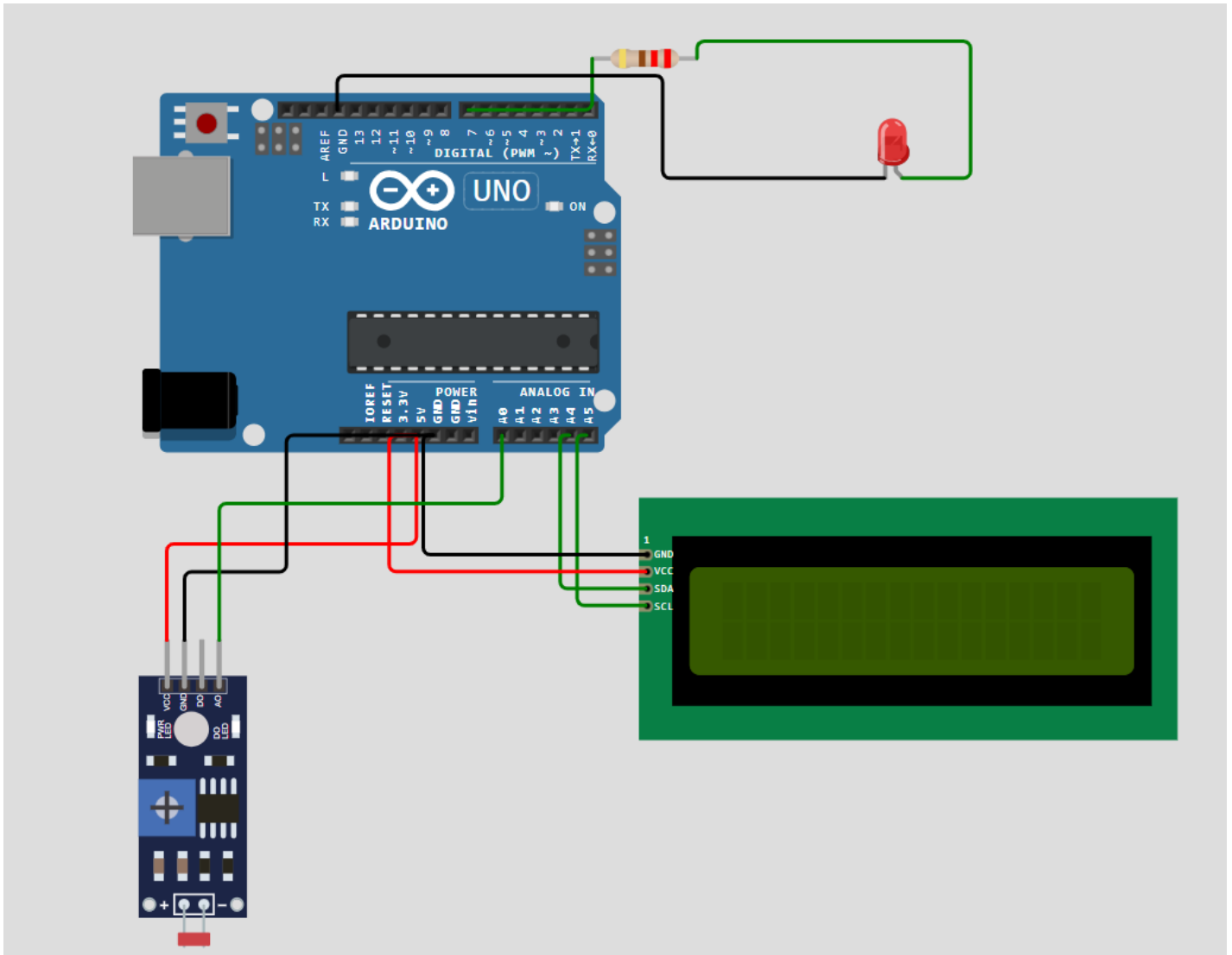


FIGURE-1 SMART LIGHTING SYSTEM USING LDR,LCD & ARDUINO UNO

WORKING PRINCIPLE

The LDR sensor continuously measures the surrounding light intensity and sends the data to the Arduino through analog pin A0.

When the light intensity falls below the predefined threshold value of 500 ($\text{Light Value} < 500$), the Arduino identifies a dark environment and turns ON the LED. The LCD displays the light value and the message "Dark - LED ON".

When the light intensity exceeds the threshold value of 500 ($\text{Light Value} > 500$), the Arduino identifies a bright environment and turns OFF the LED. The LCD displays the light value and the message "Bright-LED OFF".

SOURCE CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);
const int ldrPin=A0;
const int ledPin=7;

void setup()
{
  pinMode(ledPin,OUTPUT);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("Smart");
  lcd.setCursor(0,1);
  lcd.print("Lighting System");
  delay(2000);
  lcd.clear();
}

void loop()
{
  int lightValue=analogRead(ldrPin);

  lcd.setCursor(0,0);
  lcd.print("Light: ");
  lcd.print(lightValue);
  lcd.print(" ");

  if(lightValue<500)
  {
    digitalWrite(ledPin,HIGH);
    lcd.setCursor(0,1);
    lcd.print("Dark - LED ON ");
  }
  else
  {
    digitalWrite(ledPin,LOW);
    lcd.setCursor(0,1);
    lcd.print("Bright-LED OFF");
  }

  delay(1000);
}
```

Code Explanation:

The program uses an LDR sensor to measure the surrounding light intensity through analog pin A0. The Arduino continuously reads the sensor value and displays it on the LCD screen. If the light intensity is below the threshold value (500), the system identifies a dark environment, turns ON the LED, and displays "Dark - LED ON" on the LCD. If the light intensity is above the threshold value, the LED is turned OFF and the LCD displays "Bright-LED OFF". Thus, the system automatically controls lighting based on ambient light conditions.

RESULTS AND CONCLUSION

Results

Dark Condition

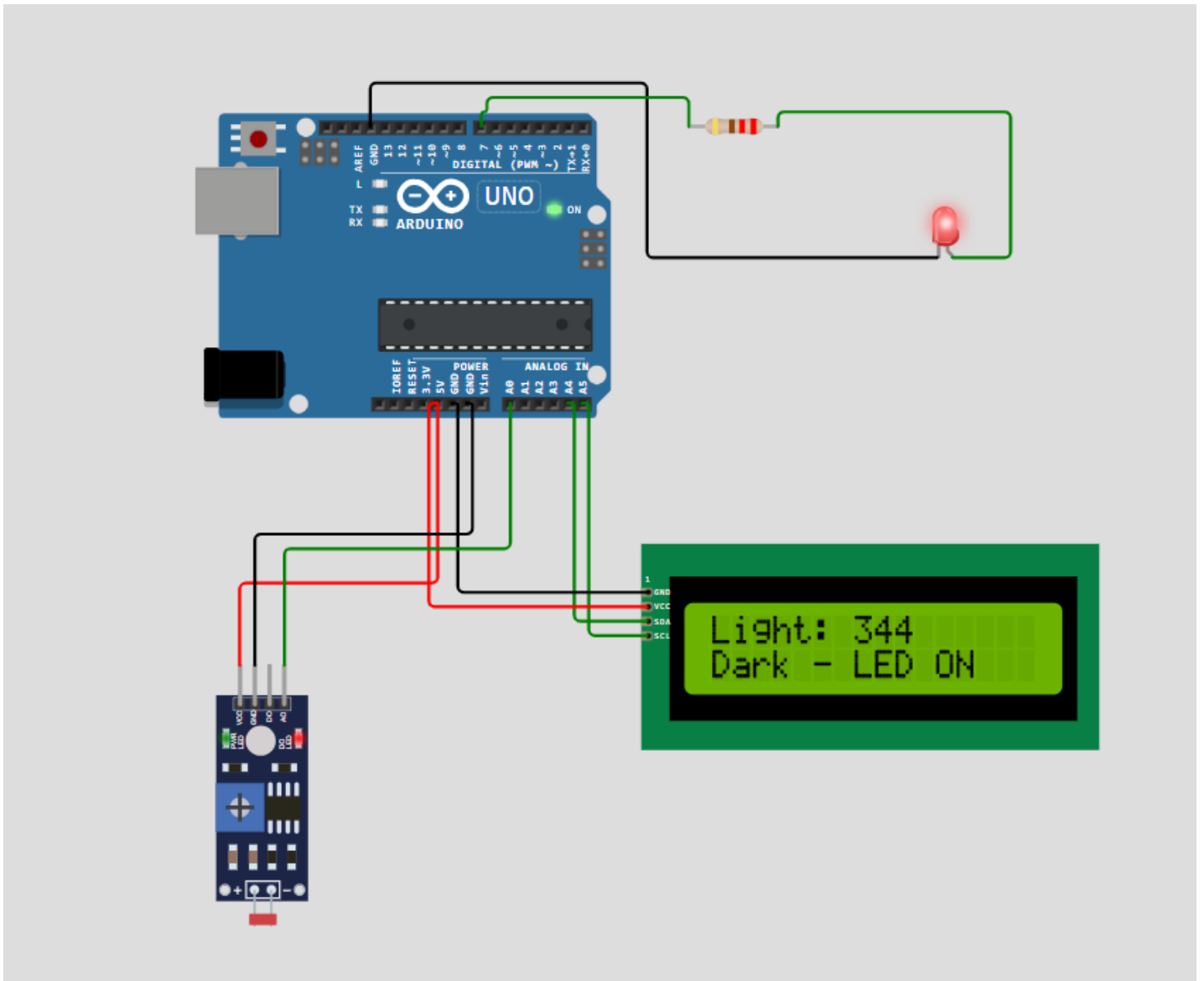


Figure 2: System operation under dark conditions. Light Value = 344 (< 500). LED Status = ON.

Bright Condition

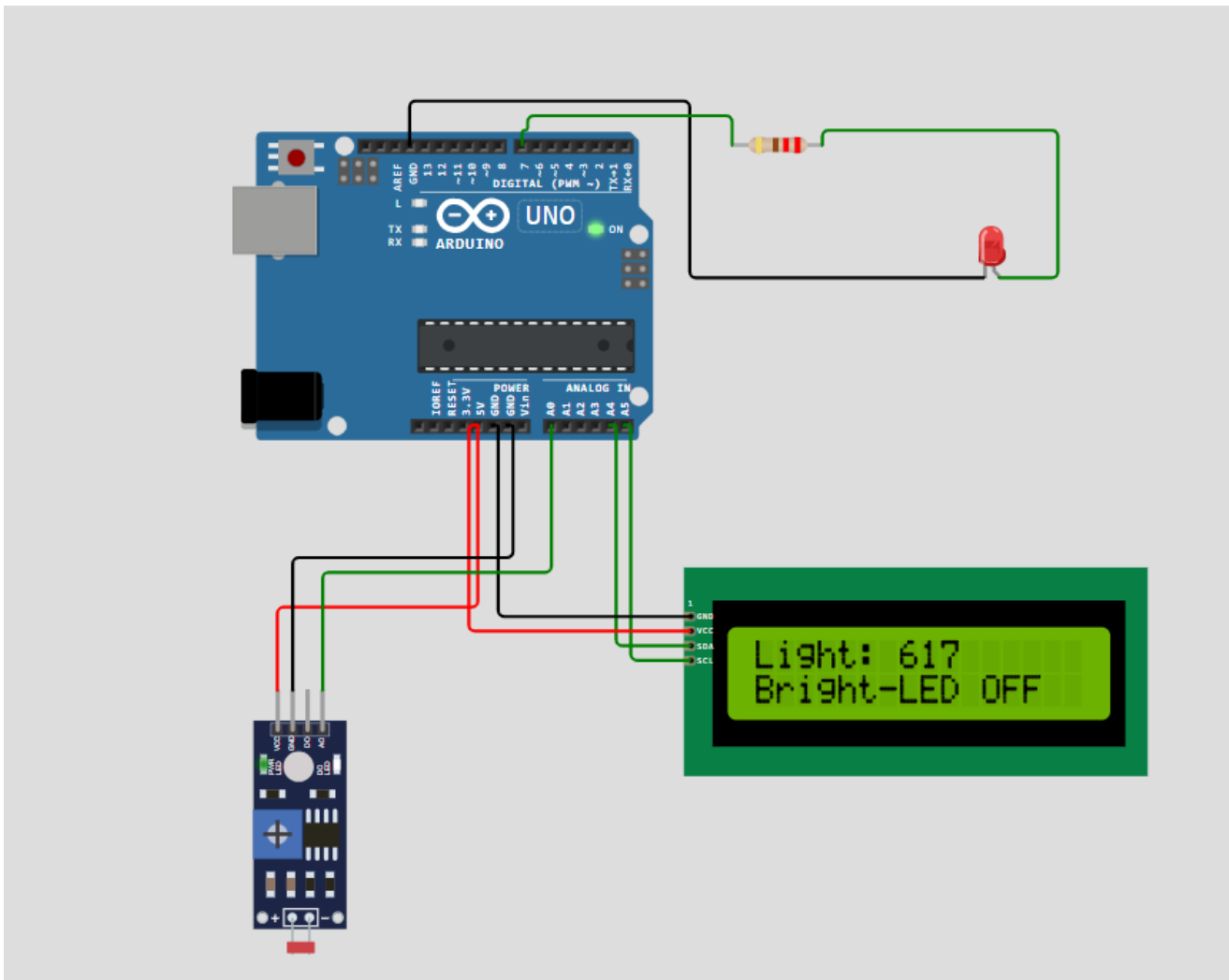


Figure 3: Figure 3: System operation under bright conditions.

Light Value = 617 (> 500)

LED Status = OFF

Applications

- Smart Street Lighting
- Home Automation
- Security Lighting
- Industrial Automation
- Energy Conservation Systems

Conclusion

The Smart Lighting System using LDR, LCD and Arduino Uno was successfully simulated in Wokwi. The system automatically controls lighting based on ambient light intensity and displays the system status on the LCD. This project demonstrates sensor interfacing, automation and real-time monitoring concepts used in Embedded Systems and IoT applications.